Bias and Variance, Overfitting, and the Support Vector Machine

5

Pre

CS 3244
Machine Learning

NUS National University of Singapore | Computing

Recap and Forecast

Linear Regression – Real valued output

$$J(\theta) = \frac{1}{m}\sum_{j=1}^{m}\left(\theta^\top x^{(j)} - y^{(j)}\right)^2$$

Logistic Regression – Probability of an event

$$J(\theta) = \frac{1}{m}\sum_{j=1}^{m} ln(1 + e^{-y^{(j)}\theta^\top x^{(j)}})$$

Gradient Descent

- Take steps in opposite the gradient of the cost function.
- SGD: the average pointwise direction is equivalent to the gradient $-\nabla J$

Learning Outcomes for this week:

- Understand the bias–variance tradeoff
- Understand why overfitting occurs: the role of model complexity and the sampled dataset
- Apply bias and variance decomposition in simple scenarios with and without noise

- Understand why maximizing the margin leads to an optimal separating hyperplane
- Understand why an SVM's model complexity is defined by its support vectors
- Extend SVMs to tolerate errors through soft margins

Bias and Variance, Illustrated

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Example: univariate sine target

$f: [-1,1] \to \mathbb{R}$

$f(x) = \sin(\pi x)$

Only two training examples! $m = 2$

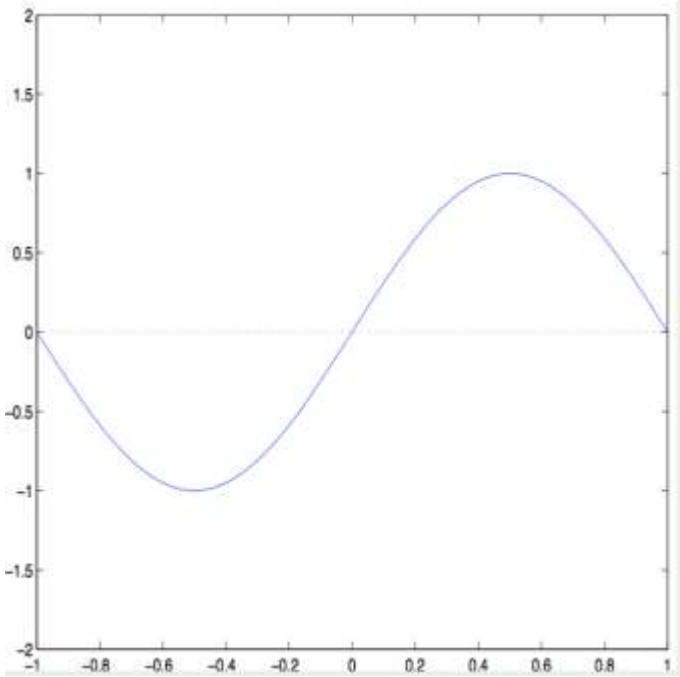
Two models used for learning:

$\mathcal{H}_0$: $h(x) = \theta_0$

$\mathcal{H}_1$: $h(x) = \theta_1 x + \theta_0$

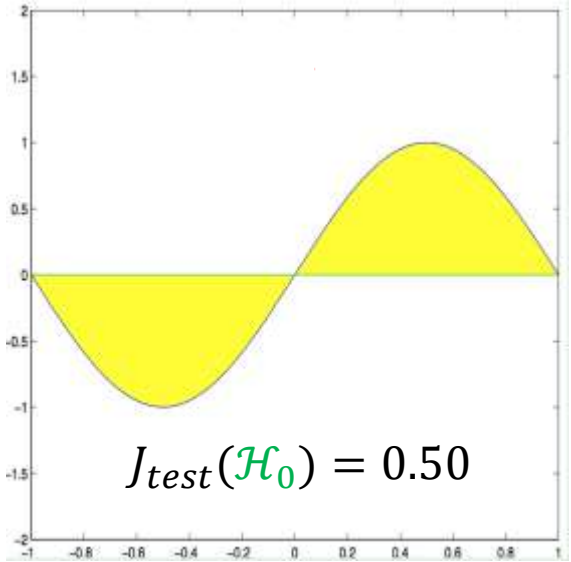
Which is better, $\mathcal{H}_0$ or $\mathcal{H}_1$?

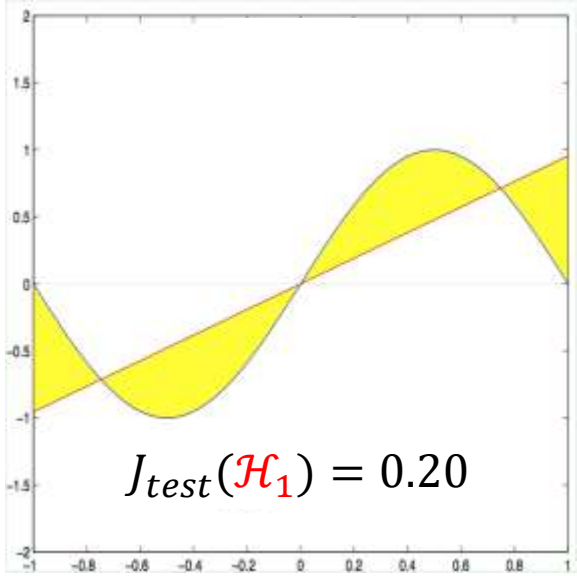
Better for <u>what</u>?



Approximation – $\mathcal{H}_0$ versus $\mathcal{H}_1$

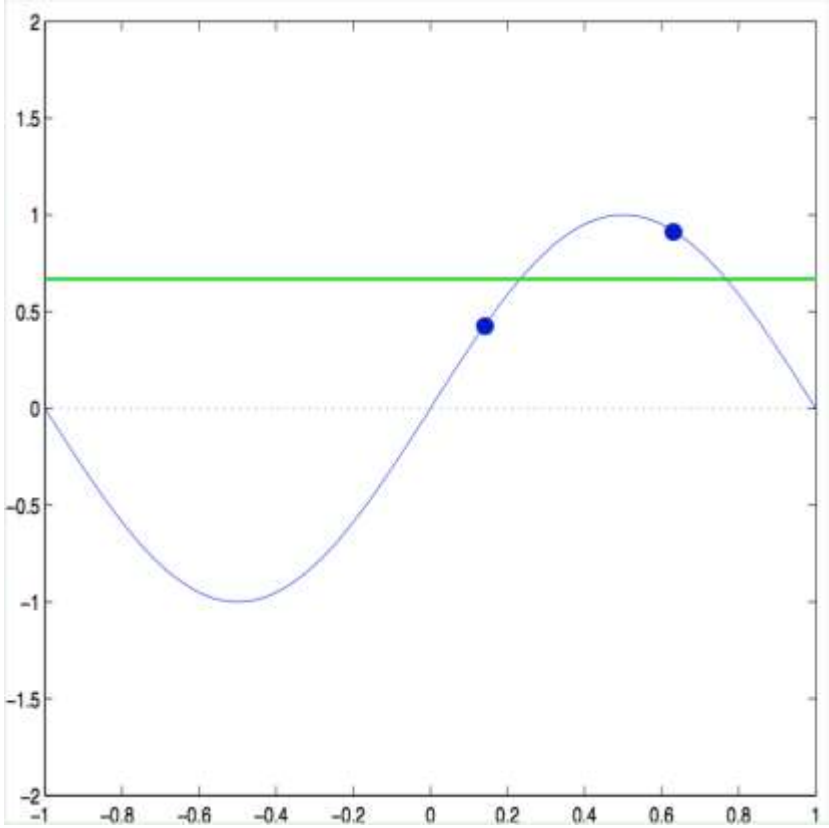
Use the full power of the model

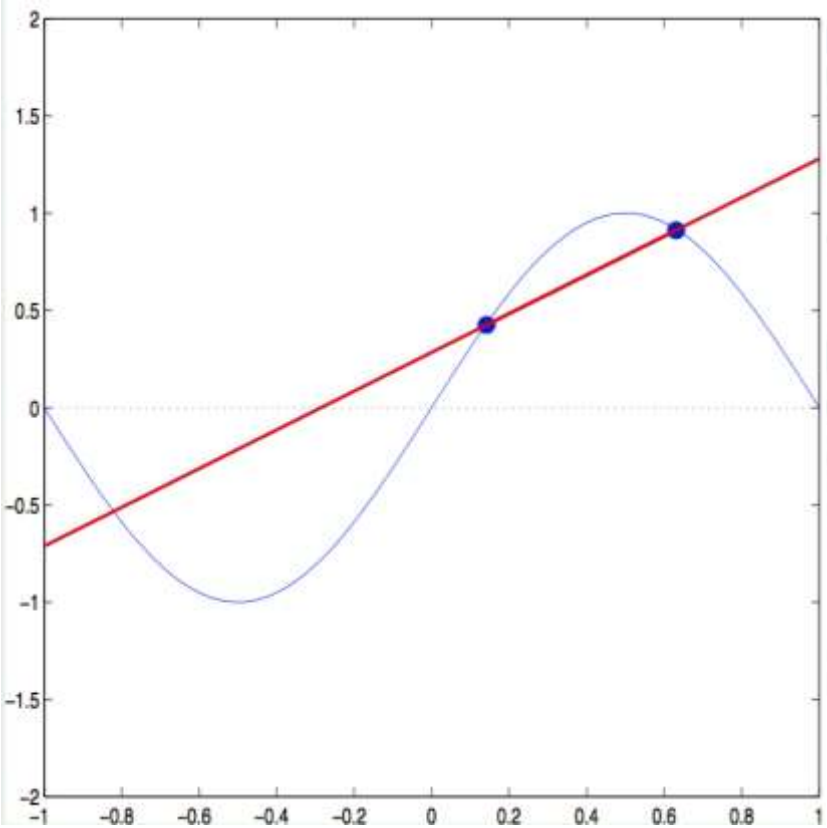




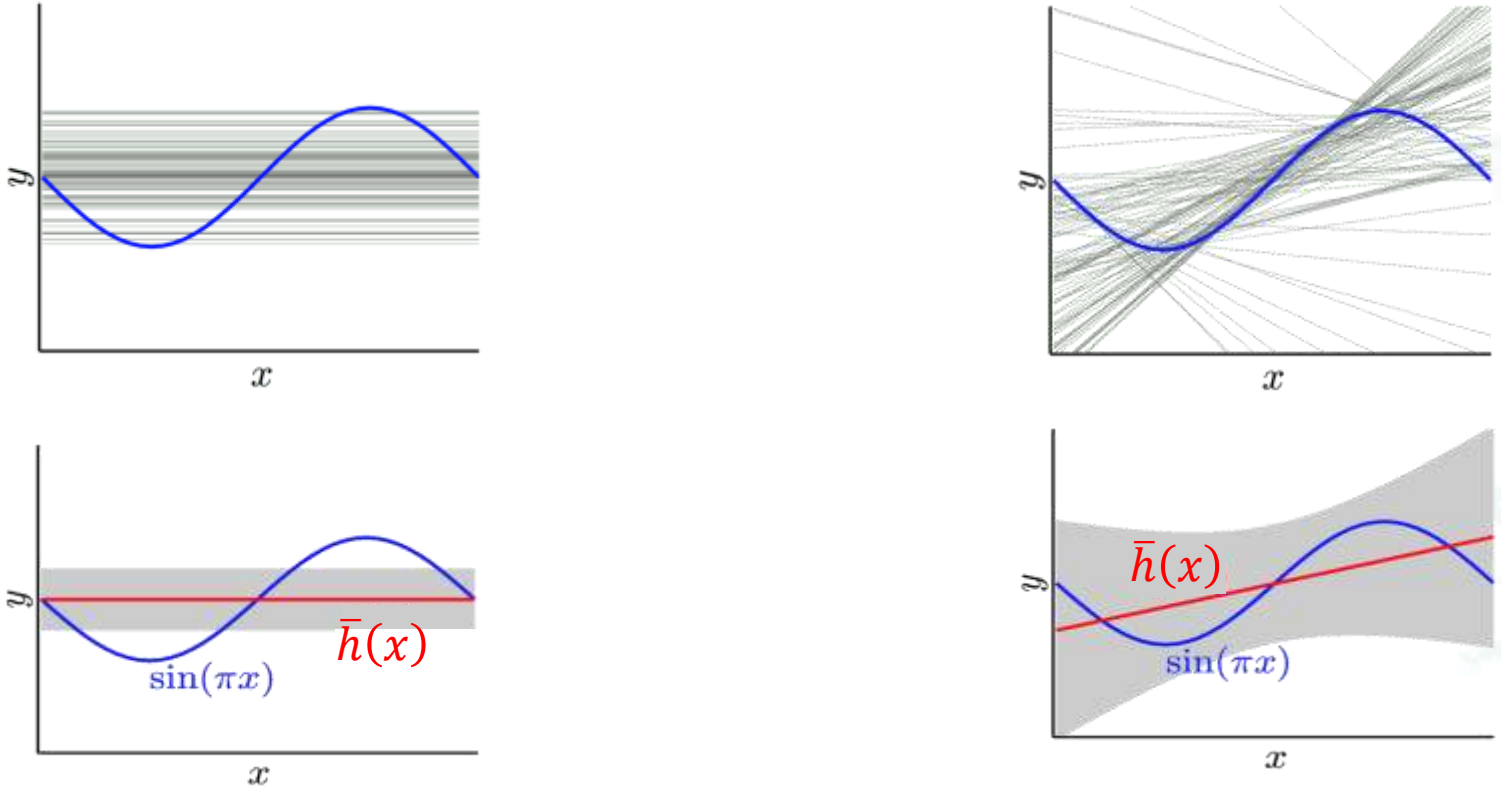
where — and — are the $\bar{h}(x)$ of each $\mathcal{H}$.

Learning – $\mathcal{H}_0$ versus $\mathcal{H}_1$



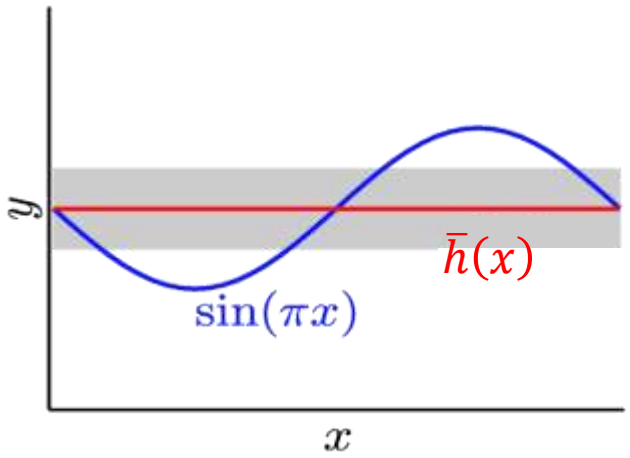


Bias & variance - $\mathcal{H}_0$ versus $\mathcal{H}_1$



And the winner is …

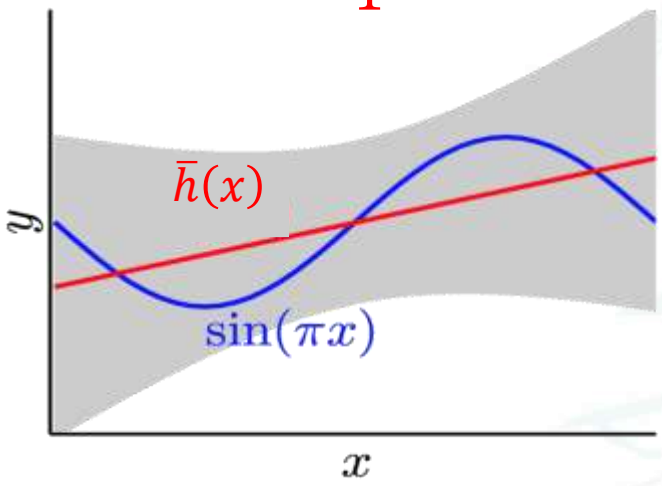
$\mathcal{H}_0$



Bias = 0.50

Var = 0.25

$\mathcal{H}_1$



Bias = 0.21

Var = 1.69

Lesson learned

Match the 'model complexity' to the

data resources, not to the **target complexity**.

Quality and Quantity

Overfitting

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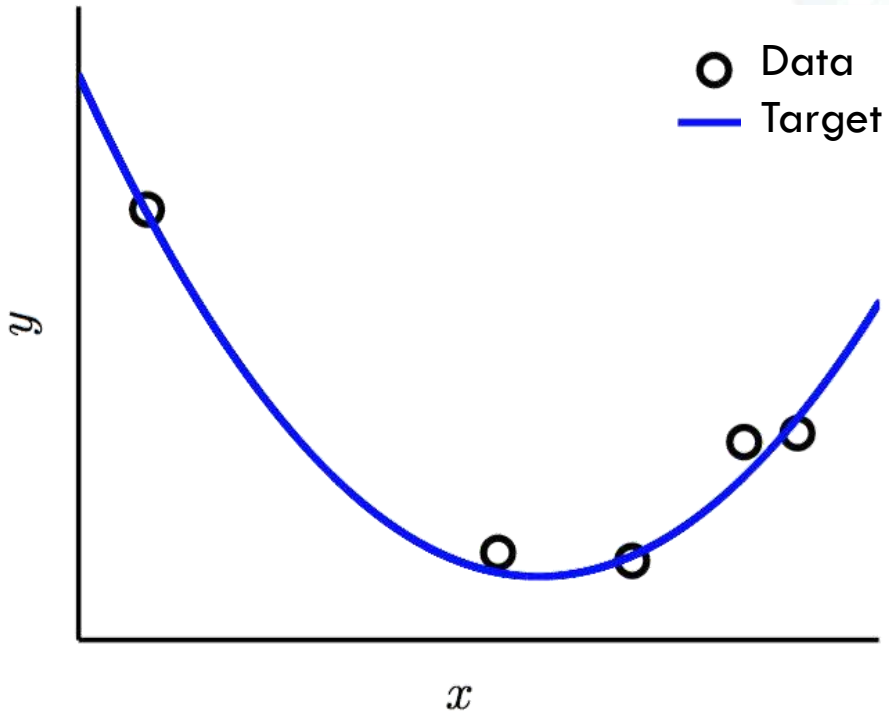
Overfitting, Illustrated

Simple target function

5 data points –
slightly noisy

4th-order polynomial fit

$J_{train} = 0$ but J_{test} is huge



Overfitting, Illustrated

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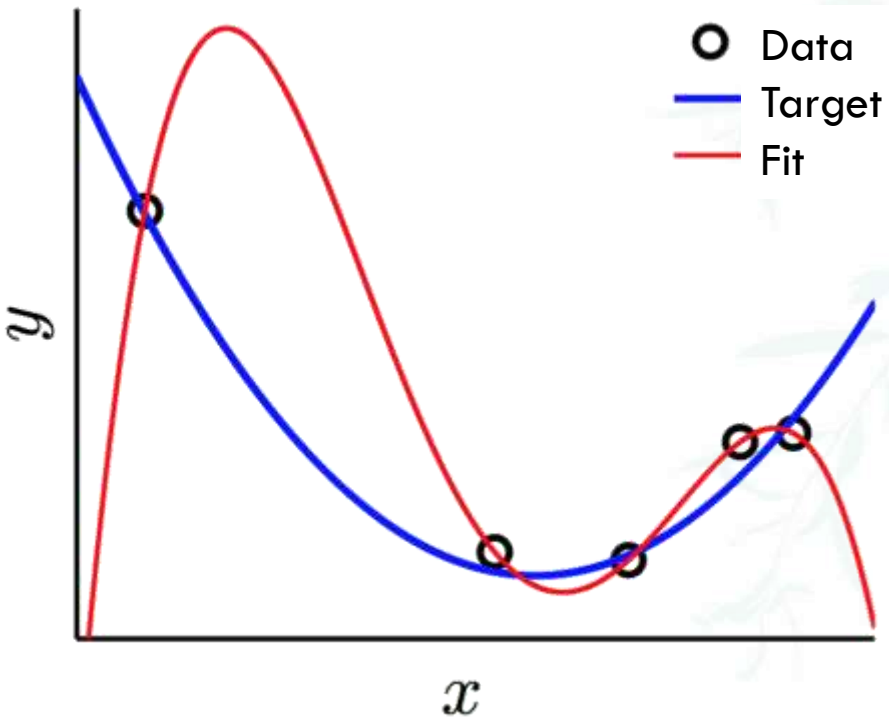


The culprit

Overfitting:
"Fitting the data more than is warranted"

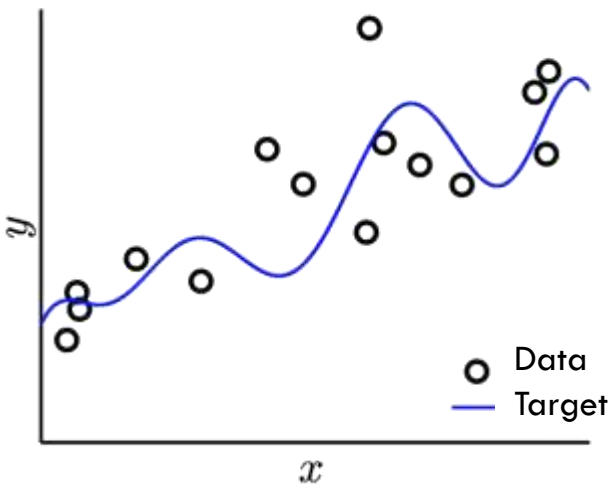
Culprit: fitting the noise.

Not just useless,

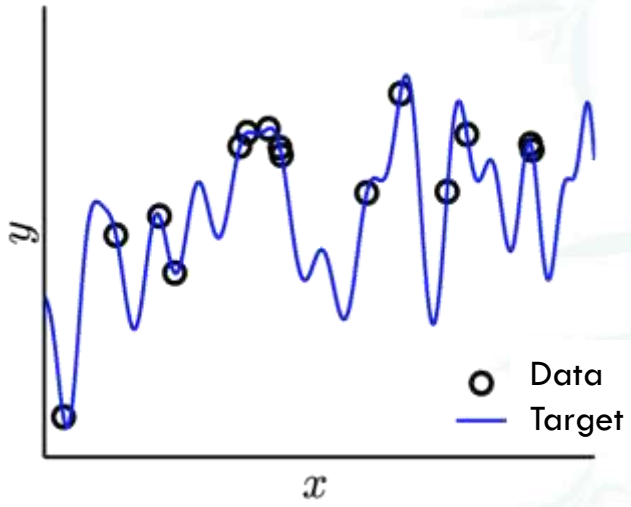
but harmful

Case study: 2nd versus 10th order polynomials

10th-order target + noise



50th order target, noiseless

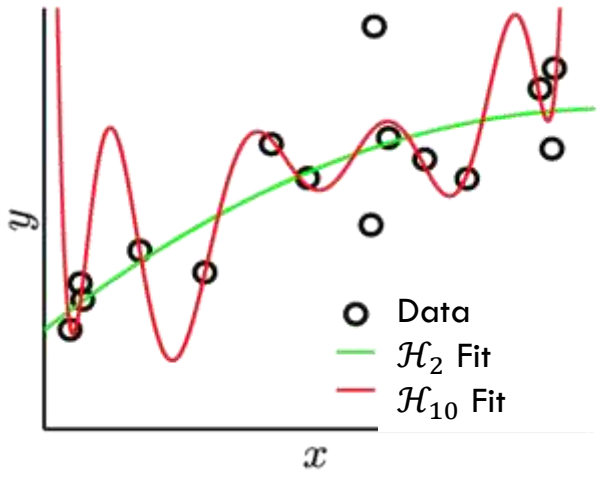


Given: 15 data points

Poll: pick your model of choice for each scenario

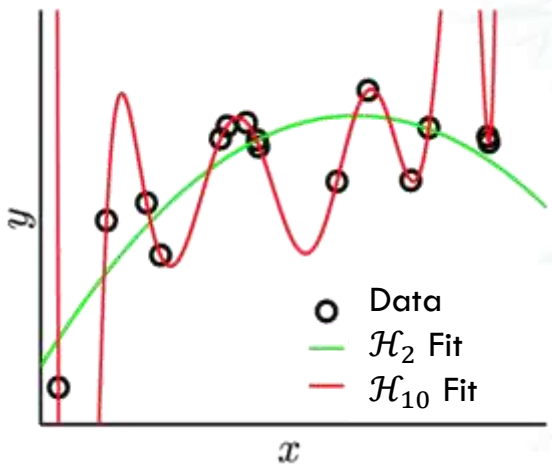
Two fits for each function

Noisy low-order target



	2nd Order	
J_{train}	0.050	
J_{test}	0.127	

Noiseless high-order target



	2nd Order	
J_{train}	0.029	
J_{test}	0.120	

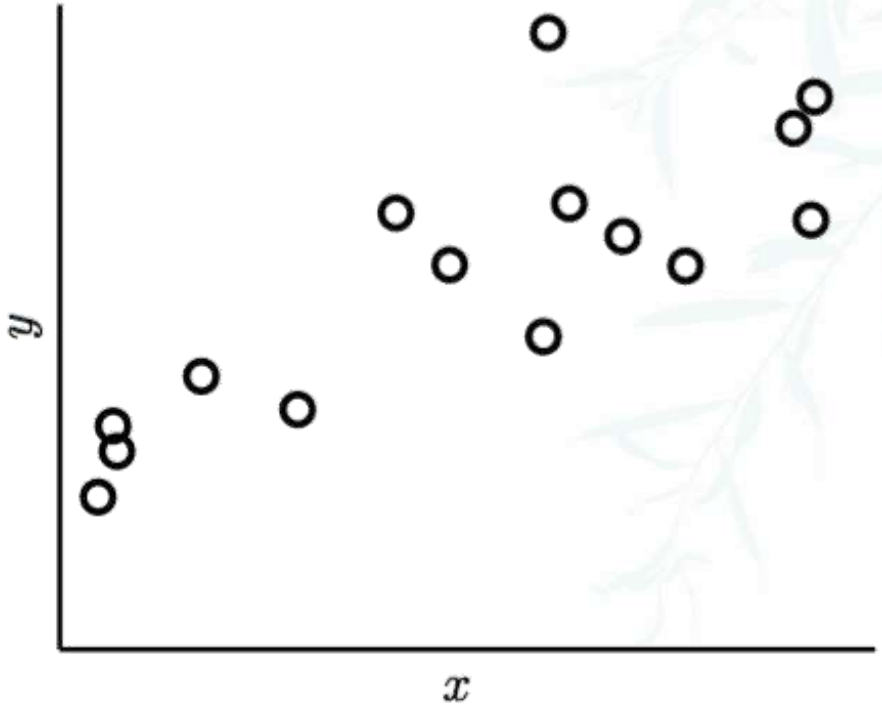
Two learners

Two learners Overfitting and Restrained:

Both <u>know</u> the target is 10th order, you get 15 data points.

O chooses $\mathcal{H}_{10}$
R chooses $\mathcal{H}_2$

Who wins?



The Support Vector Machine

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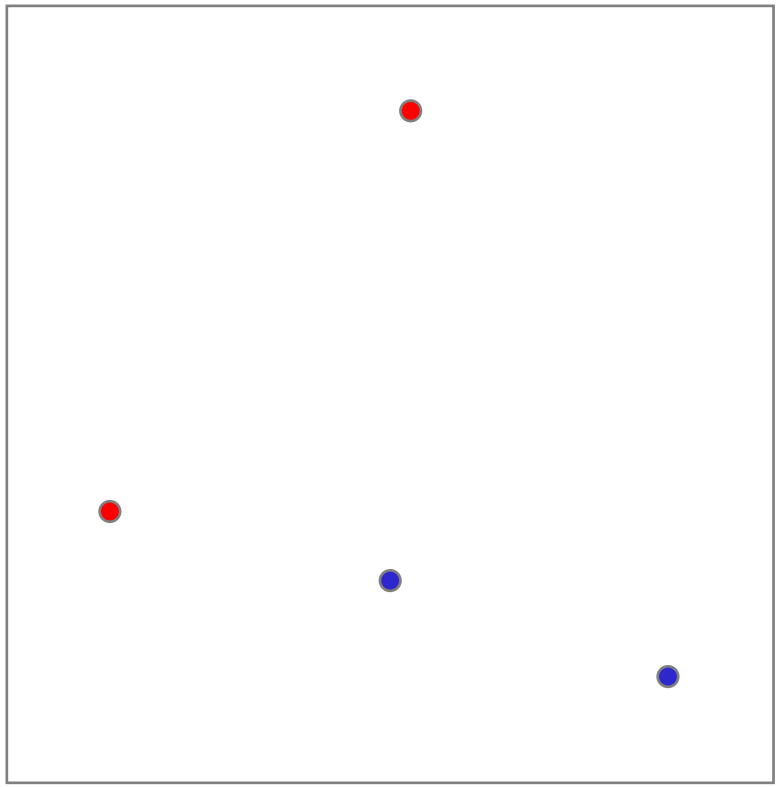
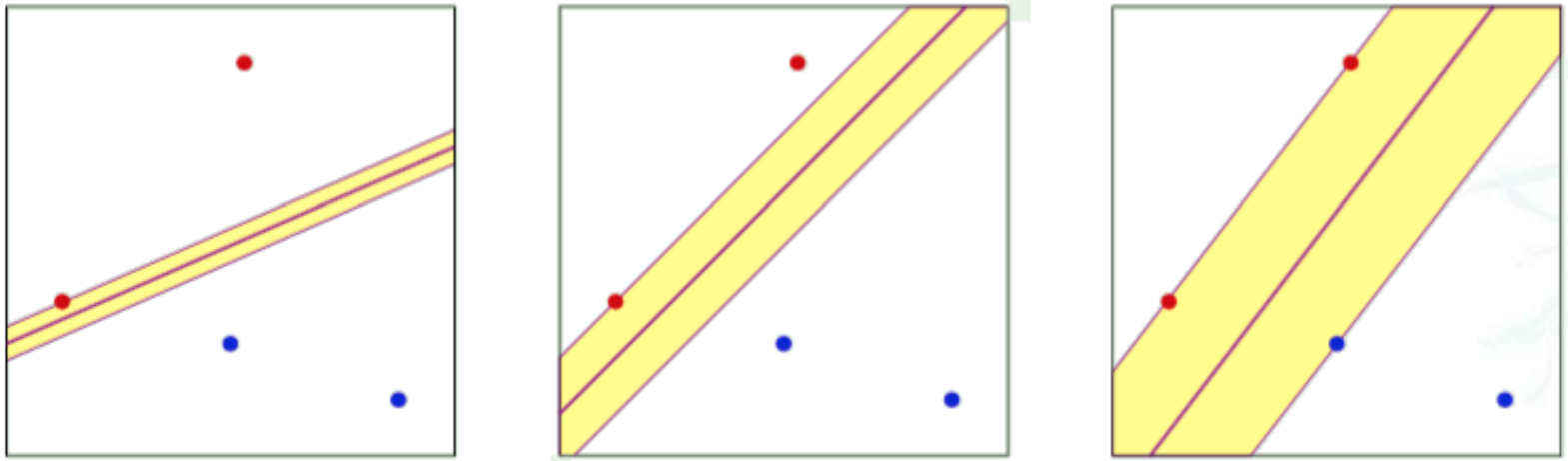
Land Transport Authority



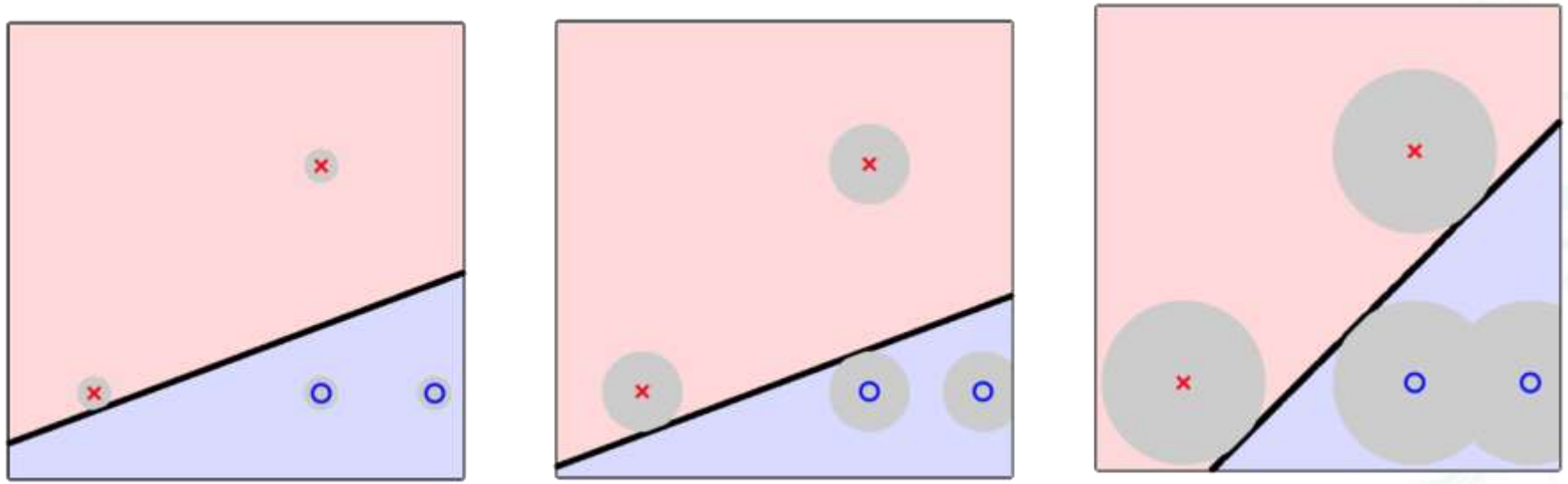
Better linear separation

Two questions:

1. Why is a bigger margin better?
2. Which θ maximizes the margin?



Inherently handles noisy data

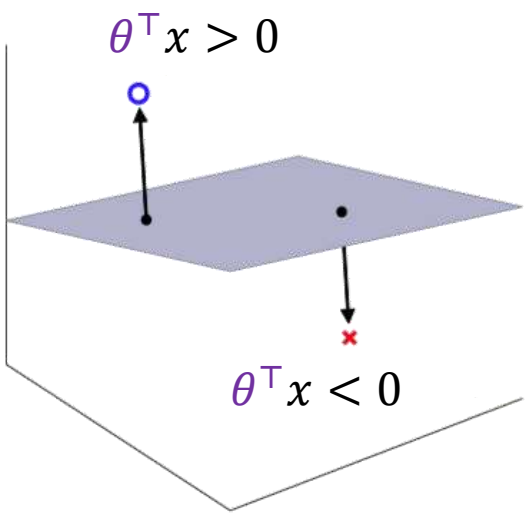


Outline of the derivation

Define the distance to the nearest points as a unit distance (1).

Solve inequality constraints using constrained optimization.

These nearest points are support vectors.



The support vectors entirely determine the classifier boundary.

The complexity of the classifier is determined by its support vectors.

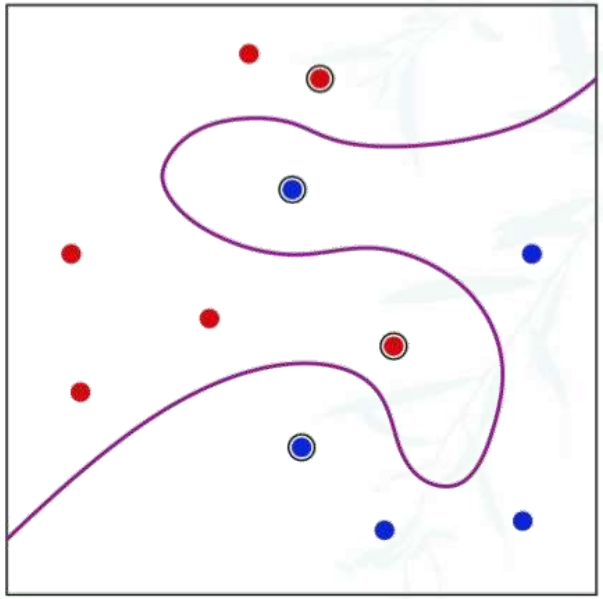
"Support Vectors" in $\mathcal{X}$ space

What if the support vectors live in $\mathcal{Z}$ space?

In $\mathcal{X}$ space, we say that we have "pre-images" of support vectors.

The margin is maintained in the $\mathcal{Z}$ space.

Great generalization, since in the model, # of parameters $\propto$ # of support vectors.



Soft Margin SVM

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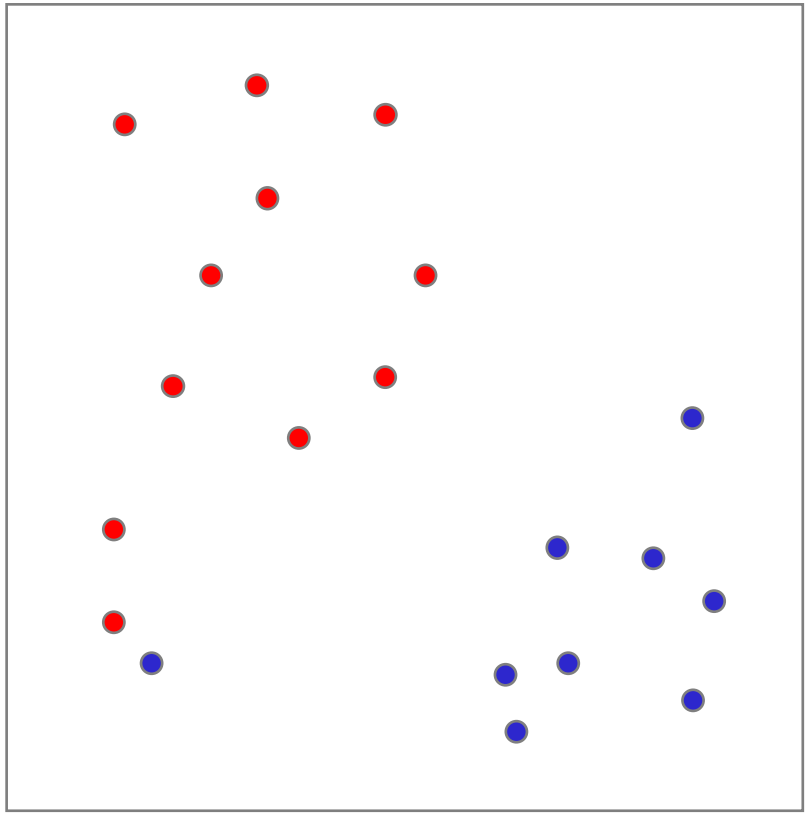
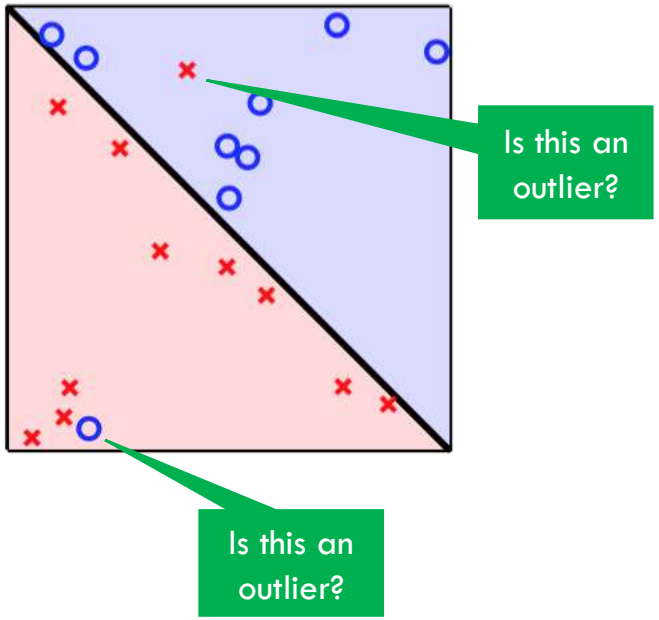
Land Transport Authority, Round 2

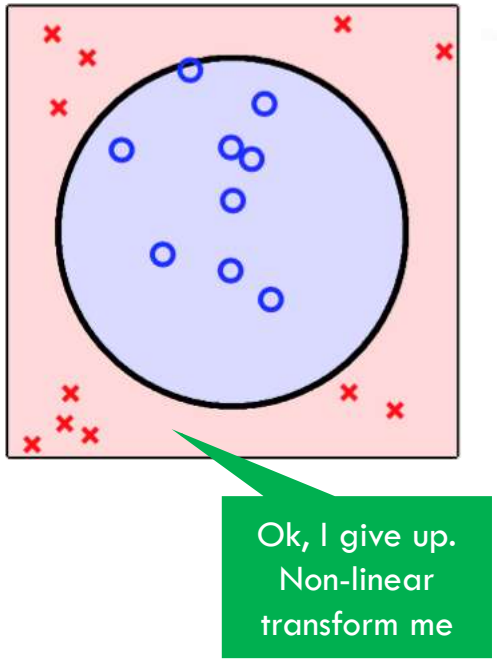


Two types of non-separable

Slightly



Seriously



Error measure

Margin violation: $y^{(*)}\left(\theta^\top x^{(*)} + b\right) \geq 1$ fails

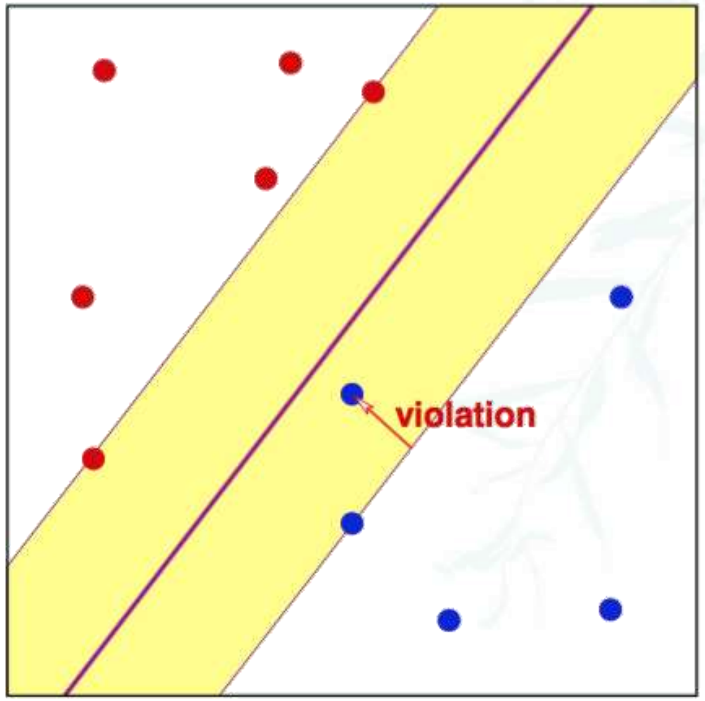
Quantify this:

$$y^{(*)}\left(\theta^\top x^{(*)} + b\right) \geq 1 - \xi^{(*)}$$

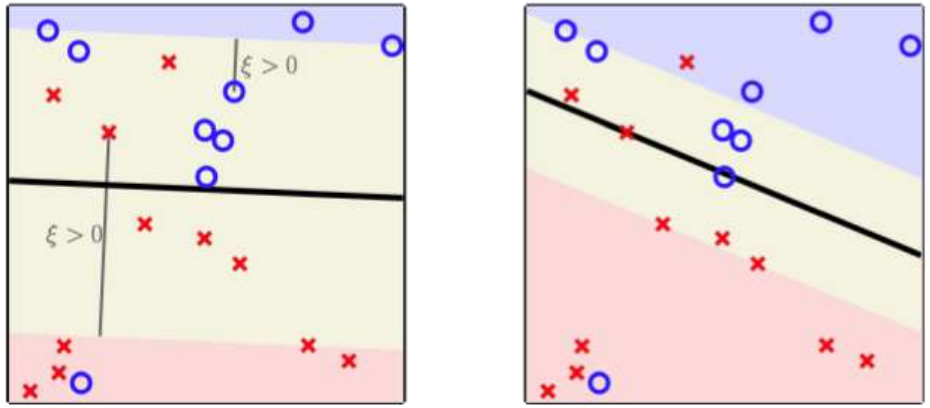
where $\xi^{(*)} \geq 0$

Slack variable: Soft error on $(x^{(*)}, y^{(*)})$

Total violation: $\sum_{j=1}^{m} \xi^{(j)}$



Soft Error C parameter

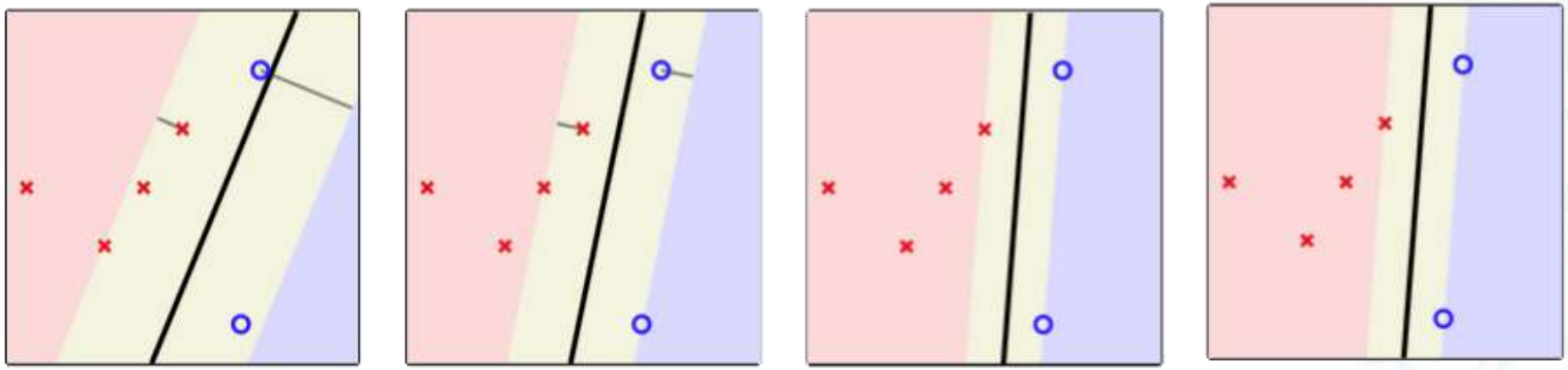


$C = 1$ $\qquad$ $C = 500$

Minimize $\quad \frac{1}{2}\theta^\top\theta + C\sum_{j=1}^{m}\xi^{(j)}$

Soft error "badness". Higher values indicate less tolerance.
Not a budget like in regularization.

Effect of Varying C



small C | medium C | large C | hard margin